



PRODUCT NAME

Advanced Eye Glass Cleaner Liquid Formula

Description

A professional-grade, streak-free eyeglass cleaner that dries quickly and maintains safety for lens coatings. Formulated to be non-toxic and ensures optimal clarity for all types of lenses.

Technical Specifications

pH Level:	7.0
Viscosity:	Low viscosity, suitable for spray applications
Shelf Life:	24 months
Batch Size:	10-100 liters
Processing Time:	2 hours
Temperature:	Ambient temperature (20-25°C)
Storage Conditions:	Store in a cool, dry place away from direct sunlight.
Equipment:	Mixing tank, Stirring device, pH meter, Quality control instruments
Certification:	GMP, ISO 9001

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Deionized Water	Aqua	78.5%	Solvent, base for the solution
2	Isopropyl Alcohol	Isopropanol	15.0%	Cleaning agent, enhances quick drying
3	Caprylyl/Capryl Glucoside	Caprylyl/Capryl Glucoside	2.5%	Surfactant, aids in streak-free cleaning
4	Sodium Citrate	Sodium Citrate	1.0%	pH buffer, stabilizes pH at desired level
5	Sodium Benzoate	Sodium Benzoate	1.0%	Preservative, ensures product shelf life
6	Glycerin	Glycerin	1.0%	Humectant, prevents drying of lenses
7	Polysorbate 20	Polysorbate 20	1.0%	Emulsifier, ensures uniform consistency
Total			100%	

Manufacturing Process

Phase 1: Mixing Phase

1. Prepare deionized water in a clean, sanitized tank.
2. Add Isopropyl Alcohol to the water under gentle stirring.
3. Slowly add Caprylyl/Capryl Glucoside and continue stirring until fully dispersed.
4. Incorporate Sodium Citrate while maintaining a controlled pH test to ensure stability.
5. Add Sodium Benzoate and mix until fully dissolved.
6. Slowly incorporate Glycerin, ensuring even distribution.
7. Finish by adding Polysorbate 20, ensuring the entire mixture is homogeneous.

Required Equipment

Mixing tank, Stirring device, pH meter, Quality control instruments

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: GMP, ISO 9001

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 10-100 liters.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Spray directly onto lens surface and wipe with a clean microfiber cloth until the lens is clean and dry. Ensure both lens sides are coated and dried evenly.